UNIVERSIDADE DA CORUÑA

Rnasa Imedir

Unit 1:

Supervised Learning

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Machine Learning I

fic facultade de informática da coruña

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Introduction

- Two phases of learning:
  - Training
    - The examples are presented to the system (training set)
    - The system “learns” from the examples
    - The system gradually modifies its adjustable parameters until the output matches the desired output (target)
    - We need a measure of performance/accuracy/error
  - Utilization
    - New examples never seen before
    - We ask the system to generalize
    - Need of a test set

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Introduction

- Generalization:
  - We do not want the system to memorize
    - Give the correct answer only to the training examples
    - Hard for humans, easy for a computer
  - We want to learn to generalize
    - Easier for humans (to conceptualize) than for a computer (to represent concepts)
      - The animal and human mind looks for patterns even where there are none
        - Pareidolias
    - We have to extract the essence, the structure of the data, and not just learn the correct answer for some cases
  - Important: test in new cases once the system is trained
    - Check if the good results are beyond the training set

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Introduction

- The dataset is made up of tuples
  - <attributes, value>
  - Training set

- We want a model or system that, when faced with new attribute values, is capable of returning a value consistent with the data set
  - Train a model
  - Generalization ability
    - To be applied to new data

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Introduction

- Training:
  - The ability of a model to solve a problem (knowledge) is linked to the examples used during learning.
  - The training set must:
    - Being meaningful:
      - Sufficient number of examples
      - If the training set is small, the network will not be able to solve the problem efficiently.
    - Be representative:
      - Various examples: all regions of the state space must be sufficiently covered.
    - If a learning set contains many more instances of one type than the rest, the model will specialize in that subset and not be of general application.

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Introduction

- If a model has been well chosen, with adequate complexity, and trained, with a well chosen training set, it will have good generalization ability
  - Will work well with new instances

- What happens if there are instances in some region where there were no data in the training?
  - Poorly chosen training set: it was not representative
    - That area where we want to use it was not represented in the training set

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Introduction

- What happens if there are instances in some region where there were no data in the training?
  - Example:
    - Train a model with 763,231,921 Asian camels with 2 humps
      - Result: the model will predict that camels have 2 humps
      - What if we use as “input” an African camel (dromedary)???
        - It will return that it has 2 humps, but it only has one!!!
      - The model was used with an instance where there were no training samples
      - Poorly chosen training set: it was not representative
        - The area where we want to use it was not fully represented in the training set

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Introduction

- What happens if there are instances in some region where there were no data in the training?
  - Example:
    - Do all cats have fur?
      - It is not so important to observe many as that all breeds are represented

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Introduction

- Parameters vs. hyperparameters
  - Parameters: Each of the values that define the behavior of a model
    - They are internal to the model.
    - Their values are set by the training process
  - Hyperparameters: Parameters whose value is used to control the learning process
    - Based on them, the values of the parameters are determined
    - Their values are set by the user

Model	Parameters	hyperparameters
Polynomial	coefficients	order
ANN	weights, bias	architecture, learning rate, activation functions, etc.
SVM	α	kernel, C, etc

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Introduction

- Problems that can be solved in supervised learning:
  - Prediction
  - Curve fitting
  - Classification
  - Regression

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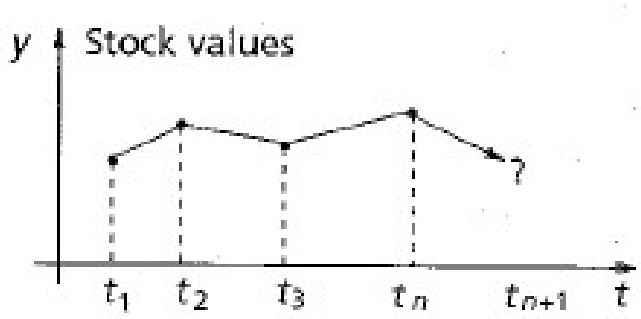
Introduction

- Applications:
  - Prediction
    - It attempts to determine the function that maps a set of input variables to one (or more) output variables.
    - It is basically numerical
    - It is based on statistical assumptions
    - Examples:
      - Monitoring the reservation of seats in aviation companies.
      - Short-term financial predictions



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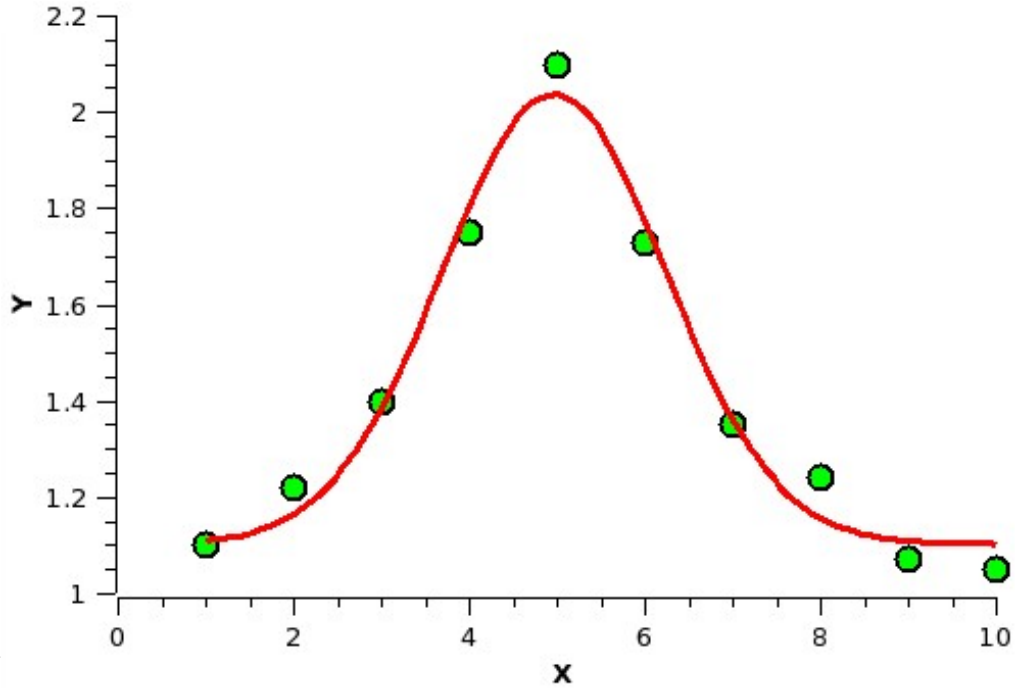
Introduction

- Applications:
  - Curve fitting:
    - Model the curve that lies behind a set of examples (xi , y)



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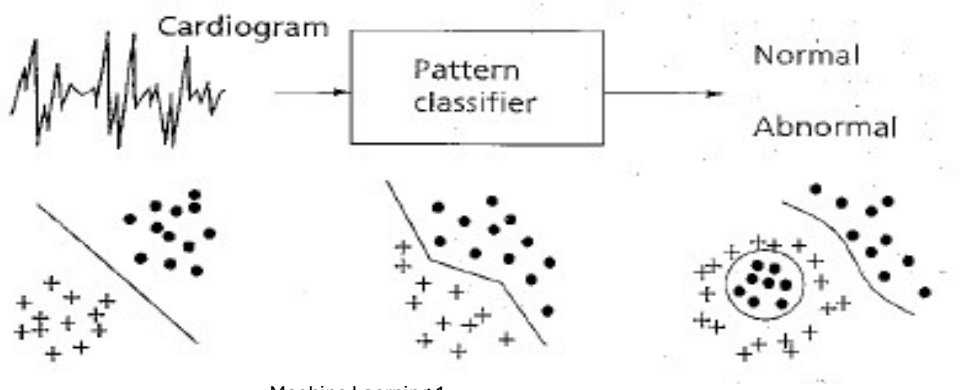
Introduction

- Applications:
  - Classification
    - Classify objects into a finite number of classes, given their properties (inputs)
    - Look for a mapping function that allows class 1 to be separated from class 2 and class 3 from class 3…
    - The number of classes is finite and known
    - The class membership of each sample is known



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Introduction

- Applications:
  - Regression:
    - Generic
    - From a dataset, try to find a model of the relationship between two sets of them
      - Instances: (inputs, targets)
      - Unknown relationship
      - Model must be generalizable, that is, that from new unknown samples it returns coherent outputs
    - The other problems (prediction, curve fitting) can be seen as particular cases of this
    - To assess the goodness of a regression system, some type of error (mean error, ECM, etc.) is used

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Introduction

- Supervised classification:
  - Encoding of the output if the model does not return categorical outputs (eg: ANNs), 2 possibilities:
    - As real numbers
      - Only if in the real world there is an order between the equivalent classes
        - Example: classify the quality of a product as low/medium/high
          - Outputs 0/0.5/1 or 0/1/2 ...
        - The classification problem becomes a regression problem
    - As boolean values (much more common)
      - More than 2 classes (one-hot encoding)
      - 2 classes (A/-A, -/+, A/B, etc.)

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Introduction

- Supervised classification:
  - Output encoding: As boolean values
    - More than 2 classes (one-hot encoding):
      - One output per class is used
        - Target: each target takes the value 1 if that instance belongs to that class, or 0 otherwise
        - Output: Real values:
          - ANN: The softmax function is applied to the outputs of the last layer (output yi):

$$\hat{y_i} = \frac{e^{y_i}}{\sum_{j=1}^{C} e^{y_j}}$$

            - From real values, it returns values between 0 and 1
              - The sum of all is equal to 1
            - Each value: probability of belonging to that class
              - Final output: class with higher certainty/probability

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Introduction

- Supervised classification:
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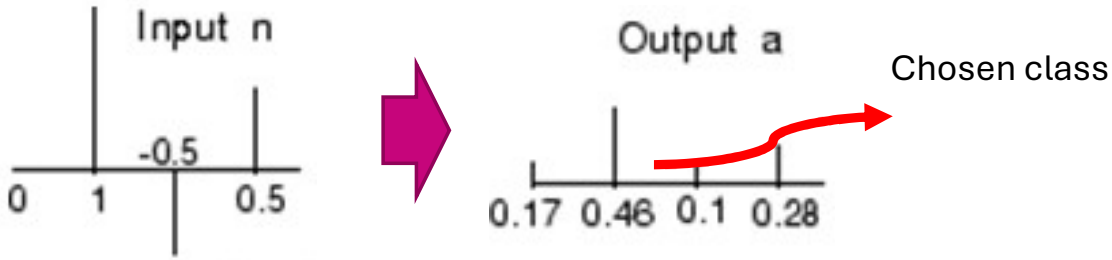
    - More than 2 classes (one-hot encoding):
      - One output per class is used
        - Target: each target takes the value 1 if that instance belongs to that class, or 0 otherwise
        - Output: Real values:
          - ANN: Example of softmax



4 outputs: real values

4 outputs: values in [0,1], sum = 1

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Introduction

- Supervised classification:
  - Output encoding: As boolean values
    - More than 2 classes (one-hot encoding):
      - ANN: Error calculation:
        - cross-entropy:
          - Given two distributions p(x), with the true values, and q(x) with the estimates, the entropy is calculated for a set of instances x in as

$$H(p,q) = -\sum_{x} p(x) \cdot \log(q(x))$$

          - Given the output of an ANN for an example n, the targets $y = p(x)$, and the outputs $\hat{y} = q(x)$:

$$H(y,\hat{y}) = -\sum_{n=1}^{N} y_n \cdot \log(\widehat{y_n})$$

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Introduction

- Supervised classification:
  - Output encoding: As boolean values
    - More than 2 classes (one-hot encoding):
      - ANN: Error calculation:
        - cross-entropy:

$$H(y,\hat{y}) = -\sum_{n=1}^{N} y_n \cdot \log(\widehat{y_n})$$

- $y_n \cdot \log(\widehat{y_n})$ is the dot product of a vector of c components (c classes):

$$y_n \cdot \log(\widehat{y_n}) = \sum_{c=1}^{C} y_{nc} \log(\widehat{y_{nc}})$$

  - Since there is only one target equal to 1, the rest are 0, only the output $\widehat{y_n}$ corresponding to the positive target is evaluated:
- Averaging across all examples:

$$J = -\frac{1}{N}\sum_{n=1}^{N} \log(\widehat{y_n})$$

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Introduction

- Supervised classification:
  - Output encoding: As boolean values
    - 2 classes (A/-A, -/+, A/B, etc.):
      - The target is encoded as 0/1 or -1/1 values
        - Depends on the classifier
        - For example:
          - ANN: 0/1
          - SVM: -1/1

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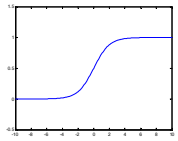
Introduction

- Supervised classification:
  - Output encoding: As boolean values
    - 2 classes (A/-A, -/+, A/B, etc.):
      - The model returns real numbers as output
        - In addition to the classification value, it gives information on the safety or certainty of the classification
        - ANNs
          - Targets: 0/1
          - The sigmoid function is usually applied as transfer function
            - High values: output tends to 1
              - Greater certainty of classification as positive
            - Low values: output tends to 0
              - Greater certainty of classification as negative
          - Classification: Threshold in 0.5

$$f(n) = \frac{1}{1+e^{-n}}$$



      - Error calculation ($\widehat{y_n}$: target, y_n: output):
        - Difference between output and target
          - ME, MSE, etc.
        - Binary cross-entropy:

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Introduction

- Supervised classification:
  - Output encoding: As boolean values
    - 2 classes (A/-A, -/+, A/B, etc.):
      - The model returns real numbers as output
        - In addition to the classification value, it gives information on the safety or certainty of the classification
        - SVM
          - Targets: -1/1
            - Output tends to ∞: Higher certainty of clas. as positive
            - Output tends to -∞: Higher certainty of clas. as negative
          - Classification:
            - Threshold in 0
              - Look at the sign

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Introduction

- Supervised classification:
  - Output encoding: As boolean values
    - 2 classes (A/-A, -/+, A/B, etc.):
      - ANN: Error calculation:
        - ($\widehat{y_n}$: target, y_n: output)
        - Difference between output and target
          - ME, MSE, etc.

$$\sum_{n=1}^{N} |\widehat{y_n} - y_n|$$

        - Binary cross-entropy:

$$-\frac{1}{N}\sum_{n=1}^{N}[\widehat{y_n}\log(y_n) + (1-\widehat{y_n})\log(1-y_n)]$$

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Introduction

- Supervised classification:
  - Many models only allow binary classification
    - Two classes
    - For example Support Vector Machines
  - To be able to apply them to multi-class problems:
    - Voting based on the combination of binary classifiers
      - “One-vs-All”
      - “One-vs-One”

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Introduction

- Supervised classification:
  - One-vs-All or One-vs-Rest:
    - Consider the problem as a collection of binary problems (2 classes)
    - For k classes, k binary classifiers are trained (k>2)
      - one per class
      - i-th classifier (i=1,…,k):
        - Data belonging to the class i → “positive”
        - Rest of the data → “negative”
      - Similar to one-hot-encoding
    - To select the class to which each data belongs
      - The sample is used as input to the k classifiers
      - A voting scheme is usually used among the k classifiers

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Introduction

- Supervised classification:
  - One-vs-All or One-vs-Rest:
    - Usual voting scheme: winner-takes-all
      - Let fi (x,w) be the output of classifier i
        - Numerical output
      - The label of the classifier that has a greater certainty that the data (x) is in the +1 class is assigned, that is

$$f(x,w) = \arg\max_i (f_i(x,w))$$

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Introduction

- Supervised classification:
  - One-vs-One:
    - In this method k(k-1)/2 models are built separating each class from each other
      - That is, all possible pairs of one class vs. another are created, and these problems are solved, in total they are k(k-1)/2
      - For each pair (i,j), i < j, of class labels, a binary classifier is constructed, classifying the points of class i with the label +1 and those of class j with the label -1
    - Subsequently, a voting method is used to select the class of each data

$$\hat{k} = \underset{k}{\operatorname{argmin}} \frac{\sum_{l=1}^{L} |m_{kl}| g(m_{kl}, s_l)}{\sum_{l=1}^{L} |m_{kl}|}$$

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